# Sahil **Kureel**

## About me

Atal Bihari Vajpayee - Indian Institute of Information Technology & Management, Gwalior - ( B.TECH + M.TECH ) In Information Technology
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**Programming Languages**

- Python 3+ • Java 17+
- JavaScript (ES6+) • SQL
- C / C++ • Bash

**Frameworks & Libraries**

`Frameworks:`
- FastAPI | Flask | Django (Python)
- NodeJs | ReactJs | ExpressJs | Tailwind (Javascript)
- spring-boot | Maven (Java)

`Databases and Storages:` PostgreSQL / AuroraDB ( RDS ) | DynamoDB | S3 / GCS / Firehose | MongoDB | Redis / Memcached
`Cloud Tech:` AWS | GCP | LocalStack | OpenTofu | Terraform
`Messaging Systems:` RabbitMQ | Kafka | SQS
`Tools:` Docker | Jenkins-pipline | SonarQube | Git
`Concepts:` Linux Systems | Distributed Systems | System Designs at High Level as well as Low Level | Performance Optimization & Tuning for scale | Data Structures & Algorithms | REST API Designs Principle (SOLID) | Concurrency Control using Mutex / Semaphores for Multithreading and Asynchronous Programming | Encryption at REST using AES-CBC / AES-GCM | CI/CD piplines.

**AI Data Tools and Tech**

- `Ollama / GitHub Copilot / GPT-4 / Claude Haiku 4.5`
- `generative AI / Retrieval-Augmented Generation (RAG)`
- `DynamoDB-Streams / PySpark / Apache-Airflow`
- `NumPy / Pandas / SciPy`
- `LLMs Integration with Ollama & FastAPI As Serving Interface`

## Experiences

Mar'26 - present | **HandShake AI**
AI Implementation Engineer · Bengaluru
**AI Coding-Agent Evaluation and RAG Platform | Project Dynamo, Terminal Bench 2, Sentinel Ultra**
**Tech:** RAG, LLMs, Python3, FastAPI, Django, NodeJs/ExpressJs, Java/Spring Boot, DynamoDB, RDS, Docker, ReactJs, Tailwind CSS.
**Project Dynamo**, an AI engineering initiative for creating repository-level tasks used in **Terminal-Bench-2** to evaluate coding agents in realistic terminal environments.
**Insights:** Built and maintained benchmark-oriented software projects spanning Python/FastAPI/Django, Node.js/Express and Java/Spring Boot, including repository scaffolding, application setup, tooling and test workflows. Developed AI-assisted workflows involving RAG applications and LLM-based engineering tasks, enabling AI systems to retrieve project context and perform multi-step software-engineering operations.

Aug'21 – Sept'25 | **HEAD DIGITAL WORKS PVT. LTD.**
SDE - III ( Platform Team ) · Bengaluru
**Unified Payment Platform | Distributed Payment Processing**
**Tech:** Java 21, Spring Boot, AWS - ECS, Lambda, API Gateway, DynamoDB, RDS, Kafka, Docker
**Scale:** 100M transactions/day, 1M TPM
**Impact:** 100M transactions/day and 1M TPM, with an architecture designed for horizontal scaling and provider-independent integration.
**Insights:** Built a unified abstraction over payment aggregators such as Juspay and Razorpay, while also supporting KYC and bank-account validation. The key engineering challenge was making the platform reliable despite external payment-provider latency, retries and asynchronous callbacks. Separated synchronous API operations from asynchronous processing using RabbitMQ/Lambda and used DynamoDB/event streams for scalable event processing. Owned production reliability and performance issues, rather than only implementing individual APIs.

**Distributed Financial Reconciliation Engine | Financial Infrastructure**
**Tech:** Python3, Java17, AWS - DynamoDB Streams, Lambda/ECS, RabbitMQ, PostgreSQL ( AuroraDB-AWS ), Docker.
**Scale:** 500K+ transactions/day, 2,000+ concurrent requests/sec.
**Impact:** 99.9% financial discrepancy detection, 15+ hours/week manual effort saved.
**Insights:** Designed a distributed reconciliation engine that continuously consumed transaction changes and compared records across multiple ledger systems. Instead of doing reconciliation synchronously, you used DynamoDB Streams + Lambda + asynchronous workers to distribute the workload. You added concurrency control, retries, scheduled jobs, audit reporting and transaction-status tracking. The system automatically flagged 99.9% of discrepancies and removed more than 15 hours of weekly manual reconciliation work.

Aug'20 – July'21 | **GARG CONSULTANCY CO.**
Software Engineer - Backend · Gurugram
**Tech:** Python(Flask/Django), NodeJs(ExpressJs), AWS, PostgreSQL, RabbitMQ, Docker
**Insights:**
- Led backend development for financial and digital platforms using Python and NodeJs.
- Designed and implemented RESTful APIs, CMS systems and backend services with scalability and security best practices.
- Collaborated with product and clients to deliver production-ready solutions.
- Mentored junior developers and improved engineering workflows.
- Managed cloud deployments and backend infrastructure on AWS.
- Supported frontend integrations with ReactJS & NextJs applications.

## Certificates & Grants

**2016** SnakDown Qualifier Round-3, CodeChef Coding Competition.
**2017** Management Team ( Technical Assistant ) at ACM - ICPC, Gwalior region.
**2019** AWS Certified Solution Architect Associate.